

Mangrove swamps

On the fringes of warm tropical oceans, salt marshes are replaced by swampy forests of salt-tolerant trees called mangroves. These tidal forests extend along more than 60 percent of tropical shorelines, where they help stop coastal erosion and flooding by intense tropical storms. They also provide food and shelter for an amazing variety of wildlife.

▼ SEEDLING SPEARS

Seeds cannot sprout in the airless tidal mud, so mangroves hang on to their seeds until they turn into tiny seedling trees. These then drop off, usually at high tide so they float away to other shores. Each seedling has a long, sharp root that stabs into the mud where it settles, enabling it to grow into a new tree.



BREATHING ROOTS

Tidal mud is full of plant nutrients, but it is also salty, waterlogged, and airless. Most plants cannot grow in this tidal mud because they need to absorb oxygen through their roots. Mangroves can survive thanks to exposed roots that absorb air through breathing pores. Some have roots that stick up from the mud as a mass of spikes. Others have aerial roots that sprout from high up the trunk and arch down through the air into the stagnant mud.

▼ FLOODED FOREST

At high tide most of the mangrove swamp is flooded with seawater. Small fish swim in from the sea to feed among the tangled roots of the trees, which also provide protection from bigger predators.

